EE-170L Computer Programming Lab

Switch / break/continue statements

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1 Objective

The students will learn the following topics during this lab session.

1. Understand and use the switch statement in C++.
2. Apply the break statement to control program flow.
3. Use the continue statement inside loops.
4. Differentiate between break and continue.
5. Develop simple menu-driven programs.

Switch Statement:

- The switch statement is used to select one block of code from multiple options based on the value of a variable.
- It is usually used as an alternative to multiple if-else statements when checking a single variable.
- Switch statement falls in the category of selection statements like if and else.
- The switch selection statement performs one of many different actions, depending on the value of an integer expression.
- The switch selection statement is called a multiple-selection statement because it selects among many different actions (or groups of actions)
- Each action is associated with the value of a constant integral expression (i.e., any combination of character and integer constants that evaluates to a constant integer value).

Syntax:

```
switch(expression)
{
    case value1:
        // code
        break;

    case value2:
        // code
        break;

    default:
        // code
}
```

2.1 The break Statement

- Notice the break statements in each case.
- The case statements show the program where to start executing in the block.
- The break statements show the program where to stop
- Without the break statements, the program would execute all of the lines from the matching case statement to the end of the block.

2.2 The default section

- The default section is an optional section that comes after all the case statements. The program branches to this section if none of the case expressions match the switch expression.

2.3 It is important to note that:

- The expression provided in the switch should result in a constant value otherwise it would not be valid.
- Duplicate case values are not allowed.
- The default statement is optional.Even if the switch case statement do not have a default statement, it would run without any problem.
- The break statement is used inside the switch to terminate a statement sequence.
- When a break statement is reached, the switch terminates, and the flow of control jumps to the next line following the switch statement.
- The break statement is optional. If omitted, execution will continue on into the next case. The flow of control will fall through to subsequent cases until a break is reached.

Simple Calculator Using Switch

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1, num2;
    char op;

    cout << "Enter two numbers: ";
    cin >> num1 >> num2;

    cout << "Enter operator (+, -, *, /): ";
    cin >> op;

    switch(op)
    {
        case '+':
            cout << "Result = " << num1 + num2;
            break;

        case '-':
            cout << "Result = " << num1 - num2;
            break;

        case '*':
            cout << "Result = " << num1 * num2;
            break;

        case '/':
            cout << "Result = " << num1 / num2;
            break;

        default:
            cout << "Invalid operator!";
    }

    return 0;
}
```

Tasks

1. **Write a C++ program that takes an input of a day's number and uses a switch statement to output the corresponding day's name. Here's how the program will proceed:**

   The program prompts the user to enter the day's number.

   The user enters a number between 1 and 7, where 1 is Monday and 7 is Sunday.

The program uses a switch statement to check which day was entered.

Depending on the entered number, the corresponding day's name is outputted

using the cout statement.

The break statement is used to end each case statement and prevent the program

from executing subsequent case statements.

Finally, the program returns 0 to indicate successful execution.

2. **Write a C++ program using switch that takes a character grade (A, B, C, D, F) from the user and prints:**

   A → Excellent
   B → Very Good
   C → Good
   D → Pass
   F → Fail

   If any other character is entered, display "Invalid Grade".

3. **Write a C++ program using switch that displays the following menu:**

   1 → Convert Kilometers to Meters
   2 → Convert Meters to Centimeters
   3 → Convert Kilograms to Grams
   4 → Convert Celsius to Fahrenheit

   The program should:

   Take user choice

   Ask for input value

   Perform the selected conversion

   Display the result

   Show "Invalid choice" if input is wrong